

TASHKENT INTERNATIONAL UNIVERSITY · FACULTY OF
ECONOMICS Synthesis and review

Week 15 · Project presentations, integrated review and final exam

Tashkent International University · Faculty of Economics

2025/2026 academic
year



Last week of classes

What fourteen weeks have given

217 formulas · 9 learning outcomes · two real-world data sets · one semester project.

No new mandatory methods this week.

Three tasks of the week

1. Oral defense of your project.
2. Putting 15 weeks into one map.
3. Preparing for the final exam — and knowing exactly what the exam will ask.

Review — not rereading

Rereading a syllabus from beginning to end is the most ineffective method. It gives a feeling of familiarity, not knowledge.

The entire structure of this week is built on memorization: answering with a closed book, then checking.

One optional addition

The ANOVA session is a continuation of the “three tests, one question” idea from week 13. It is **OPTIONAL** and **NOT** included in the exam. Why — we will show you at the end, and the reason is interesting.

Today's Goals

By the end of the week, you will be able to

1. Present your analysis in five minutes — in the order of question, effect, interval, condition, what was not proven — (CLO9);
2. Relate each week of the course to the learning outcome and formula range (CLO1-CLO9);
3. Describe the structure of the final exam and what is expected of each section;
4. Identify which week is weak through a self-test and make a plan for revision;
5. (optional) Relate ANOVA to a t -test and show why its conditions are violated in this data.

No new formulas this week

The course formula family stops at week 15:

**Week 1 T1-T10 Weeks 2-7
F1-F88 Weeks 9-13
F89-F201 Week 14
F202-F207 No new formulas
at week 15**

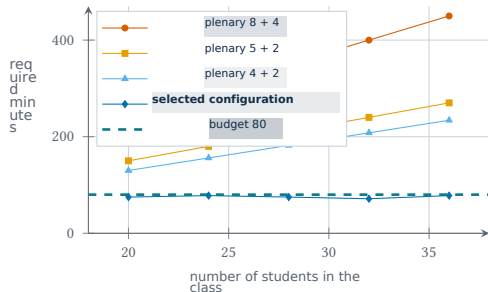
Total 217. The task this week is to reduce them, not increase them.

Common mistake

Final exam in week 16: 3 hours, 240 points, 35% of the course grade. Coverage — the entire course, but the center of gravity is weeks 9-14.

Project presentations

Presentation: first arithmetic, then format



Readability

The three plenary lines do not fall below the budget line anywhere. The line below — parallel streams — falls, but the configuration changes depending on the class size: 20 and 24 people → 2 streams, 28-36 people → 3 streams, and at 24, 32, 36 the format drops from 5 + 2 to 4 + 2. The rule "always 3 streams, 5 + 2" does not work: at 36 people it gives 90.0 minutes.

Plenary presentation is NOT POSSIBLE — measured

Number of (format, class) pairs that fit into an 80-minute content budget: 0.

format 20 28 36

8 + 4	250	350	450
5 + 2	150	210	270
4 + 2	130	182	234
3 + 1	90	126	162

Even a 3 + 1 "lightning" format takes 90 minutes in a class of 20.

Audience question

If we give all four weeks of the week in the 8+4 format (320 minutes of content), only 25 students will make it — and there will be no revision or exam practice.

Solution: parallel streams

Format (for 28 students)

number of streams 3
10 students per stream
presentation 5 minutes
Q&A 2 minutes
exchange 0.5 minutes

total per stream 75.0 minutes
budget 80 minutes reserve 5.0
minutes

For other class sizes (each can hold 80 minutes): $20 \rightarrow 2$ streams, $5 + 2$, $75.0 \cdot 24 \rightarrow 2$ streams, $4 + 2$, $78.0 \cdot 32 \rightarrow 3$ streams, $4 + 2$, $71.5 \cdot 36 \rightarrow 3$ streams, $4 + 2$, 78.0 .
The format changes with the class — increasing the number of streams alone is not enough.

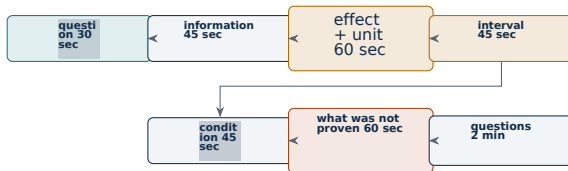
Who is the evaluator? The teacher cannot stay in each stream permanently — he rotates through the streams (an average of 25.0 minutes per stream in a class of 28 students).

So the main evaluator is the listening students. This is not a disadvantage: peer evaluation has already been practiced in weeks 12 and 14, and evaluating someone else's work improves your own.

Therefore

The peer evaluation sheet is mandatory and is included in the final grade. An unfilled sheet is equivalent to not participating in the presentation.

What to say in five minutes



skipping this step is the most common mistake

What NOT to say × Code. No one wants to read R code on a screen. × Literature review. This is 5 minutes, not a dissertation. × Every graph. One graph — the strongest one — is enough. × The “not enough time” excuse. It’s obvious that there isn’t enough time.

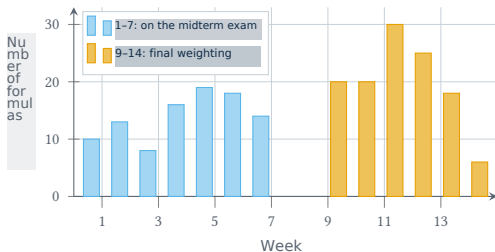
One exercise

Say your statement out loud once in advance and time it. Almost everyone gets 8–9 minutes on their first try.

You did this in 2 minutes in Week 14's lab — 5 minutes is too much time today, not too little.

Repetition map

Fifteen weeks in one picture



Coverage

Weeks 1-7: 7 weeks, 98 formulas (45.2 percent) — this was asked on the midterm exam.

Weeks 9-14: 6 weeks, 119 formulas (54.8 percent) — the center of gravity of the final exam.

But be careful: the final exam covers the entire course. You can't just "pass and forget" 1-7.

Blind Spots — Measured

Four learning outcomes occur in only one week:

CLO Topic Week

CLO1 Descriptive Statistics 1
CLO4 Correlation 6 CLO6
Confidence Intervals 9 CLO9
Communication 14

They are the least repeated — and that is why they need special time for review. The most common, CLO7 (4 weeks), usually comes to mind on its own.

Six bridges of the course — the backbone of repetition

What does the bridge say

1 4 → Week 7: $\sqrt{n}$ law The dispersion of the sample mean $\sigma/\sqrt{n}$ — four times as much information halves the error 2 7 → 9: σ is unknown If σ is replaced by s , the distribution goes from σ to s 3 9 → 10: interval = test If the value of σ_0 is not in the 95th percentile, $\sigma < 0.05$ 4 10 → 11: from one sample to two Same logic, different SE; design (paired or related- you) determines the outcome 5 11 → 12: “how strong” → “how” Correlation gives the strength of the association, regression form 6 12 → 13 → 14: estimate → cause → report σ_1 — sample estimate; causal meaning comes from design; confidence comes from transparency

Theorem Repetition method. Write each bridge in your own words with a closed book. Remember the connections, not the individual formulas. Exam Section B asks for the connections.

Common mistake

If you can't explain one bridge, reread the two weeks that relate to it — 6 bridges cover 12 weeks.

Final exam

Exam structure — open arithmetic

Number of sections points total minutes

A — short questions 18 4 72 54.0 B — problems
4 30 120 90.0 C — interpretation of information
1 48 48 36.0

TOTAL 240 180

Review

$72 + 120 + 48 = 240$ · shares 30/50/20 percent · all three sections are divided by the whole score (18×4 , 4×30).

Speed: 1.3333 points/minute. In the midterm exam, $150/120 = 1.2500$ — the ratio is 1.0667, which is almost the same. The exam is longer, not more rigorous.

What each section asks

A (72 points). Short, clear answer: definition, condition, one calculation, one interpretation. 3 minutes each.

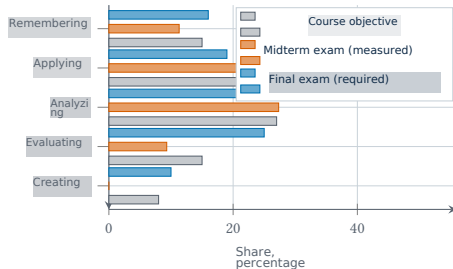
B (120 points). Full question: model → calculation → condition check → interpretation. 22.5 minutes each. The last step brings the largest part of the score.

C (48 points). You are given a ready-made analysis — with numbers — and you evaluate it: what is wrong, what is missing, is the conclusion justified.

Common mistakes

0.75 minutes = 1 point, the same for all three sections. So divide your time by the score: if you spend more than 3 minutes on a 4-point question, you are stealing from somewhere else.

Why there is a section C — and why it is not a math question



What the midterm exam showed

Measured in week 8: Applying 52.1 percent (target 35), Evaluating 9.3 percent (target 15), Creating 0.

So the midterm exam was biased toward “calculate.” It wasn’t hidden — it was measured and announced.

The final exam fixes this

When the entire grading system (6 components) is taken into account, the final exam should have the following distribution:

Recall 16 percent 38.4 points
Application 19 percent 45.6 points
Analysis 30 percent 72.0 points
Evaluation 25 percent 60.0 points
Creation 10 percent 24.0 points

Audience question

Result: The largest deviation at the course level falls from 17.1 percentage points to 0.2 percentage points. This is the only reason for section C.

WHAT IS NOT IN THE EXAM — save your time

Not required

× ANOVA — optional session today. × Maximum Likelihood Estimate (MLE). × Nonparametric tests. × Multivariate Regression — only shown today, relevant to ECMT301. × R or Python syntax by heart. Code output may be provided — it must be readable, not written. × Memorize table values — tables of $\Phi(\cdot)$ and $\phi(\cdot)$ are provided.

Allowed Formula sheet provided (with tables of $\Phi(\cdot)$ and $\phi(\cdot)$) · non-programmed calculator · ruler.

Four steps repeated in every B problem

1. Which method and why (one sentence). 2. Calculation — show each step. 3. Condition check: which condition, how was it checked, whether it was fulfilled. 4. Interpretation: effect + unit + interval + ϕ , in that order — and what was not proven.

If step 4 is skipped, the largest block of points in the problem is lost. This is the rule for weeks 12-14 and it also applies to the exam.

ANOVA — optional

ANOVA: Expanding on the idea of week 13

σ^2 and μ — DIFFERENT test in two groups

Wheat yield in Uzbekistan and Tajikistan, $\sigma^2_1 = \sigma^2_2 = 27$:

σ^2 (equal variance) 5.76086146 σ^2_1
33.18752471 σ^2 (one-factor
ANOVA) 33.18752471

$ F - t^2 $	0
$ p_t - p_F $	0

This is not approximately equal — it is exactly equal. ANOVA is a different way of writing the σ^2 -test in two groups.

30

$$F = 40.164543, p = 7.141826 \times 10^{-22}.$$

ANOVA says: at least one pair is different. ANOVA DOES NOT say: which pair.

And the problem of week 11 returns

5 groups $\rightarrow$ 10 pairs. Bonferroni limit $0.05/10 = 0.005000$.

Without correction $p < 0.05$: 7 pairs. With Bonferroni: 7 pairs. Missing pair: 0.

This is also a lesson: here the effects are so large that correction does not work. As in week 11 — correction only changes the results that are on the border.

Why ANOVA is not included in the exam — the reason is interesting

country SD Shapiro $\hat{p}$ normality

Uzbekistan	1.179486	0.002470	violated
Kyrgyzstan	0.277411	0.064013	fulfilled
Kazakhstan	0.259923	0.797913	fulfilled
Tajikistan	0.852792	0.004457	violated
Turkmenistan	0.808027	0.013292	violated

Common error

Largest SD / smallest SD = 4.537836. Levene test: $\hat{p} = 12.251092$, $\hat{p} = 1.754734 \times 10^{-8} \Rightarrow$ the condition of equal variances is violated.

And most importantly — independence

This is a time series. The 27 years of each country are related to each other — we measured this in week 13.

So the most important condition of ANOVA is violated in this data, and the above $\hat{p} = 7.14 \times 10^{-22}$ is unreliable.

Conclusion — and this is the whole logic of the course

ANOVA is easy to calculate. The design knowledge required to apply it correctly is beyond the scope of this course.

That is why it is optional: asking for a method on the exam that cannot be checked is like teaching students to calculate and taking them away from analysis.

Bridge to ECMT301

Second variable: the number doesn't change much, the MEANING changes

Variable 1 Variable 2

log(budget)	1.148411	1.147249	β^2
	0.466680	0.467339	β
			1.600751
			1.600090

corrected β^2	0.466461	0.466901
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Theorem $\beta = 2436$. The slope changed by 0.1011 percent — practically nothing. But the interpretation is completely different:

Variable 1: "if the budget increases by 1%, revenue increases by 1.1484%"

Variable 2: "if the year is the same, if the budget increases by 1%, revenue increases by 1.1472%"

"Other things being equal" This statement is the whole point of ECMT301. And it's not a new idea: it's the arithmetic form of the confounding adjustment idea from Week 13.

It does what we did with stratification in Week 13, with a multivariate regression formula.

And one catch — we'll intentionally show

The β -value of the "year" variable is 0.082863 — NOT significant. Yet the adjusted β^2 increases.

Reason: Any variable with adjusted $\beta^2/\beta > 1$ increases when added. Here $\beta = 1.735024$ — greater than one but less than the 1.960940 required for significance.

So "adjusted β^2 increased" is NOT a reason to add it to the model. This is where ECMT301 starts.

Conceptual test

Conceptual test — first individually, then in pairs

Question 1 A student is preparing for a final exam and has reread the outline three times. He feels prepared. This feeling:

(A) is a reliable indicator of preparation (B) is familiarity, not knowledge (C) is only deceptive in weak students

Answer: (B)

Rereading creates familiarity, and the brain interprets familiarity as knowledge. The only reliable test is answering with a closed book. Therefore, this week's self-test consists of 30 questions and is not for grades, but for diagnosis.

Question 2 In problem 1, you correctly calculated χ^2 , found the p -value, and wrote " χ^2_0 is rejected." You wrote nothing else. The problem is 30 points. How many points will you get?

(A) 30 (B) About 25 (C) Less than half

Answer: (C)

The final exam grade is 25 percent — 60 points. The bulk of it comes from the interpretation points in section B: 9.0 points for each problem. The condition check is also graded separately.

The calculation is correct, the answer is incomplete — this is the most common way to lose points on the exam.

Common mistakes

Method. 1 minute individual answer → find your neighbor who answered differently → 2 minutes convince him → vote again.

**Summar
y**

Most common mistakes students make — in repetition

Common mistake

1. “It’s enough if I memorize the formulas.” There are 217 formulas and a sheet is given. The exam asks not the formula, but which formula and why.

2. “I passed the midterm exam topics, I forgot.” The final exam covers the entire course. Weeks 1-7 98 formulas — 45.2 percent.

3. “If I read a lot — I know a lot.” Rereading gives familiarity. Memorizing — knowledge.

4. “If I write $p < 0.05$, the problem is over.” The answer is half-evaluated without checking the effect, unit, interval, and condition.

Common mistake

5. “I’ll study ANOVA too, maybe he’ll ask.” Not asked. Spend your time on section C — interpretation.

6. “The more I say in the presentation — the better.” 5 minutes. Not a lot of talking, but selective talking is appreciated.

7. “The adjusted R^2 has increased, so the model has improved.” We saw today: it increases even with an insignificant variable ($|t| > 1$ is enough).

8. “There is no calculation in section C, so it’s easy.” Section C is worth 48 points and it’s the hardest section; evaluating someone else’s analysis — that’s what the audit assignment for week 14 was designed for.

First Statistics is not about eliminating uncertainty, it is about measuring it.

Every estimate comes with its own standard error; every interval with its own confidence level; every $\square$ with its own conditions. Giving a number unconditionally is not statistics.

Second The method is chosen by the design, not the data.

Paired or unpaired · observation or experiment · confounding factor or collider. Same numbers, different design — different answer. Weeks 11 and 13 were about that.

Third Good analysis also says what it doesn't know.

Limitations · sensitivity · "what wasn't proven" · repeatability. In week 14, we measured: one question had 48 honest answers. Confidence comes from openness, not determination.

Audience Question

Next Step. Work through this week's 30-question self-test closed-book, identify your two weakest weeks from the diagnostic chart, and give them 2 hours each before the exam. Don't try to re-read the entire course—it won't work, and you won't have time.

Review essentials

► Çetinkaya-Rundel M., Hardin J. Introduction to Modern Statistics — “Chapter review” and exercises at the end of each chapter. Open access: openintro.org/book/ims ► Newbold P., Carlson W., Thorne B. Statistics for Business and Economics — Chapters 1-15, end-of-chapter problems.

About review methods — both short

► Roediger H., Karpicke J. (2006). Test-enhanced learning. *Psychological Science* 17(3), 249-255. — a classic example of why recall is more powerful than recitation. ► Dunlosky J. et al. (2013). Improving students' learning with effective learning techniques. *Psychological Science in the Public Interest* 14(1), 4-58. — 10 methods evaluated; only two received “high utility”.

Next step — ECMT301 and beyond

► Wooldridge J. Introductory Econometrics — Chapters 1-3 Introduction to multivariate regression. ► Angrist J., Pischke J.-S. Mastering 'Metrics — Causal Designs, continued from Week 13. ► Wickham H. et al. R for Data Science (2nd ed.) — r4ds.hadley.nz, open access.